



Large Language Models for chatbot applications handling sensitive information

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ABSTRACT

Recent advances in Large Language Models (LLMs) can enhance the performance of chatbot applications. However, LLMs rely on training data, which may limit their ability to answer questions based on up-to-date information. Moreover, many domains must address security and privacy threats to sensitive portions of this information. This paper presents an approach for designing high-performance and low-cost LLM-based chatbots that handle sensitive information. We used information retrieval techniques with the framework Rasa and integrated them into a retrieval-augmented generation architecture, leveraging low-cost online and offline LLMs. This setup enables up-to-date information and an additional layer of protection for internal documents, which may contain sensitive or proprietary content. Our case study considers an educational public policy in which we curated and annotated a dataset of 591 question-answer pairs written in Brazilian Portuguese extracted from institutional sources. We compared retrieval models, including BM25, E5 variants, OpenAI embeddings, MiniLM-L12-V2, and SBERT-MPNET, and the LLMs Sabiazinho-3 (specialized in Portuguese), Llama 3 (1B, 3B, 8B, and 70B), Gemma 3 (1B, 4B, and 12B), and DeepSeek-R1-Distill-Llama-70B. The evaluation relied on traditional metrics, such as ROUGE-L and BERTScore, as well as metrics computed using the LLM as a judge, including fidelity and answer relevance. The hybrid approach, integrating Sabiazinho-3 and Gemma 3 12B, achieved high performance, with a mean BERTScore of 0.758 for 96.672% of questions answered. The results reflected substantial semantic similarity, factual consistency, correctness, and relevance to nearly all reference answers.

1. Introduction

The adoption of chatbot applications has transformed how various sectors deliver information and provide support to users and stakeholders (Luo et al., 2022; Rhim et al., 2022; Singh & Namin, 2025). These applications support fields such as healthcare (Thirunavukarasu et al., 2023), education (Debets et al., 2025; Labadze et al., 2023), and public administration (Senadheera et al., 2024), automating customer service, enhancing operational efficiency, and providing faster and more accessible user experiences. In applications with high volumes of inquiries, especially those involving recurring questions or regulatory content, chat-

bots can deliver rapid and consistent responses, reducing the workload of human support teams (Bird & Lotfi, 2024).

Nowadays, Chatbot applications have evolved significantly, moving from rule-based systems with predefined questions, answers, and rigid dialogue flows (Luo et al., 2022; Rhim et al., 2022) to supervised machine learning models capable of more flexible intent classification and entity extraction (Rebelo et al., 2022). Recently, Large Language Models (LLMs) enabled the creation of chatbots with advanced capabilities in semantic understanding and natural language generation (Singh & Namin, 2025). However, despite their impressive performance, especially in larger and online versions, LLMs may still generate factually

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incorrect answers (hallucinations), which undermines their reliability in sensitive applications (Ji et al., 2023a). Misclassifications by online LLMs not only reduce response quality but also pose privacy risks when handling sensitive data. While offline LLMs may also suffer from misclassifications, they offer improved data privacy. Additionally, LLMs are limited to the information available in their training data (Cheng et al., 2024a), which restricts their ability to respond to specific, up-to-date, or confidential documents, such as institutional regulations or internal operational manuals.

To address the issue of relying on an outdated training set, the Retrieval-Augmented Generation (RAG) approach incorporates external evidence into the answer generation process (Lewis et al., 2020b). A typical RAG architecture comprises two main components: a retrieval module and a generation module (Chen et al., 2024). The retrieval module searches for relevant documents or excerpts from an indexed knowledge base, considering the user's query, typically using semantic matching techniques based on vector embeddings. The generation module uses an LLM to provide responses grounded in the retrieved content. By providing additional context, this setup can produce accurate, specific, and informative answers aligned with reference documentation. Thus, RAG systems are well-suited for domains governed by regulations or technical guidelines, where accuracy is essential.

Moreover, many applications should go beyond delivering up-to-date and accurate information and actively implement privacy protections, while maintaining a reasonable cost-effective strategy (Biswas, 2020; Pasupuleti et al., 2023; Sebastian, 2023). Organizations in domains such as healthcare, education, and public administration handle sensitive data and must take steps to prevent disclosure to third parties. To ensure compliance, they adhere to data protection regulations, including the European Union's General Data Protection Regulation (GDPR) and Brazil's General Law on the Protection of Personal Data (called LGPD), which are applicable in their respective jurisdictions. In RAG systems, developers often configure the retrieval module to access internal documents that may contain sensitive or proprietary content (Lewis et al., 2020a). These systems must apply data protection to preserve personal privacy and organizational confidentiality (Li et al., 2024).

This paper presents an approach for designing high-performance and low-cost LLM-based chatbots. We consider a chatbot to have high performance when its responses demonstrate substantial semantic similarity, factual accuracy, correctness, and relevance in comparison to reference answers. Moreover, a low-cost LLM is a model designed to minimize financial costs associated with training, deployment, and ongoing use. There is a considerable amount of research on investigating online LLMs, which are proprietary models and require introducing users to a cost by tokens consumed. However, studies have demonstrated the critical role of low-cost LLMs in supporting applications within resource-constrained environments (Monteiro Santos et al., 2025). For example, it is possible to customize low-cost LLMs as compact local and non-proprietary models for edge applications (Gogineni et al., 2025). Such models are capable of delivering high-quality responses even when deployed on devices with limited resources (e.g., smartphones) (Barros et al., 2025).

We combine Information Retrieval (IR) techniques, implemented using the Rasa framework (Kale et al., 2025), with a RAG architecture that enables the chatbot to access internal documents, which may contain sensitive or proprietary content, and generate contextualized responses. To preserve privacy, we route sensitive inputs to an offline LLM deployed in a secure environment, ensuring that confidential data remains local and protected. For non-sensitive inputs, we leverage an online LLM to enhance response generation with broader language understanding and fluency.

We conducted a case study on a national educational public policy, the Brazilian Textbook Program (called PNLD) (Silva et al., 2024; So-brinho et al., 2023; Tenório et al., 2025), in which we curated and annotated a dataset of 591 question-answer pairs extracted from institutional sources to support system evaluation. We assessed our implementations using traditional automated text similarity metrics and evaluations gen-

erated by LLMs to measure the accountability of the chatbot's responses in terms of quality, fidelity, and relevance. In our context, accountability means ensuring responsibility for model behaviors and outputs by evaluating the quality, faithfulness, and relevance of responses to meet the needs of sensitive domains such as public education policy.

Therefore, the main contributions of this paper include:

1. The development of a hybrid chatbot design approach using traditional IR and RAG techniques for institutional government documents written in Brazilian Portuguese;
2. A comparative evaluation of various embedding and low-cost online and offline LLM models, with a focus on privacy preservation, retrieval effectiveness, and response generation performance; and
3. The use of classical performance evaluation metrics, such as ROUGE (Lin, 2004) and BERTScore (Zhang et al., 2023), and LLM-based metrics, such as faithfulness and relevance, through the RAGAS framework to assess accountability (Es et al., 2024).

These contributions advance the application of Natural Language Processing (NLP) techniques in the public education sector, particularly for automating support tasks based on regulatory and policy documents. Researchers have explored hybrid RAG models, with a few explicitly targeting policy and legal contexts (Kalra et al., 2024). Others offer approaches potentially applicable to policy documents (Lee et al., 2024; Zhang & Chen, 2025). However, to the best of our knowledge, this is the first study to evaluate a range of embedding methods and low-cost online and offline LLM models, with a specific focus on privacy preservation in the context of large-scale public education policy. Moreover, our case study focuses on documents written in Brazilian Portuguese, which is a language that remains relatively underexplored in NLP research. Our contribution lies in demonstrating that even with minimal, cost-effective solutions, like open-source embedding models and LLMs adopted in a local infrastructure, it is possible to achieve significant performance (e.g., high faithfulness and correctness scores) on policy-focused question-answer tasks, offering a practical baseline for governments with limited resources.

The rest of the article is organized as follows. Section 2 reviews related work. Section 3 introduces our approach to designing high-performance, low-cost chatbots. Section 4 outlines our methodology, including corpus construction and setup. Section 5 presents and analyzes the experimental results. Finally, Section 6 concludes the article and outlines directions for future research.

2. Related work

Conversational systems evolved significantly over the past decades, from early rule-based systems (e.g., ELIZA) to modern solutions powered by deep learning and neural language models (Singh & Namin, 2025). The introduction of neural network-based architectures marked a turning point by enabling chatbots to generate more natural and contextually appropriate responses (Luo et al., 2022). More recently, LLMs have expanded chatbot capabilities, allowing coherent and informed responses across various domains (Singh & Namin, 2025). However, their limited access to up-to-date or external information has led researchers to integrate retrieval mechanisms to enhance their capabilities. As a result, the RAG architecture combines the generative strengths of LLMs with information retrieval techniques to produce responses that are not only coherent but also grounded in current and externally sourced data (Lewis et al., 2020b).

Leutz-Schmidt et al. (2025) evaluated ChatGPT and Gemini, varying the complexity of 31 workplace management questions in radiology. ChatGPT-4.0 outperformed Gemini, demonstrating the highest quality and clarity across all tested categories. Workum et al. (2025) benchmarked five LLMs (i.e., GPT-4o, GPT-4o-mini, GPT-3.5-turbo, Mistral Large 2407, and Llama 3.1 70B) on 1181 expert-level critical care multiple-choice questions and found that four models, including GPT-4o, outperformed human physicians in accuracy and consistency.

Table 1
Comparison of prior works and our approach.

Prior Work / Domain	Approach / Limitation	Our Approach
General RAG architecture (Lewis et al., 2020b)	Combines retrieval with generation to improve grounding but lacks privacy-aware mechanisms or dynamic model selection.	We adopt RAG principles while adding online/off-line LLM switching based on data sensitivity to address privacy needs.
Healthcare (hypertension self-management) (Montagna et al., 2025)	Uses a local classifier to filter sensitive inputs before online LLM calls, but complete local option reduces quality and no dynamic switching is applied.	We conduct the dynamic switching between online and offline LLMs for improved balance of quality and privacy.
Healthcare chatbot privacy (Li et al., 2024)	Relies on access control, de-identification, and prompt engineering, but approaches are static and may lack adaptability for varied inputs.	We consider embedding-based and model-level selection strategies for flexible, context-sensitive privacy control.
Privacy-preserving RAG in cloud-based retrieval (Cheng et al., 2024b)	Perturbs query embeddings and uses homomorphic encryption, but adds computational complexity and requires specialized infrastructure.	We do not implement cryptography directly, but suggest integration of such mechanisms as optional layers for stronger privacy.
Personalized advice with cryptographic privacy (Watanabe & Uchikoshi, 2025)	Combines LLMs with zero-knowledge proofs for user verification, but introduces cryptographic overhead and may limit scalability.	We highlight the possibility to adopt cryptographic methods while prioritizing architectural flexibility and practical deployment at scale.
Radiology, critical care, education, customer service, cybersecurity, ethics (Kagan et al., 2025; Leutz-Schmidt et al., 2025; van der Rijt et al., 2025; Shafee et al., 2025; Workum et al., 2025; Yoo et al., 2025)	Benchmark LLM performance in domain-specific settings, but often use high-cost proprietary models without privacy controls.	We extend the discussion to large-scale public education policy , emphasizing low-cost solutions crucial for government contexts with limited budgets, ensuring scalable and sustainable deployment despite financial constraints.

Despite their strong performance, all models occasionally produced incorrect answers, a behavior commonly attributed to hallucinations, which is a known limitation of current language models (Huang et al., 2025). Such experiments evaluated proprietary models or those with a large number of parameters, which limits their applicability in resource-constrained environments and scenarios involving sensitive information.

Similar studies on LLM-based chatbots exist in other domains, such as education (Yoo et al., 2025), customer service (Kagan et al., 2025), and cybersecurity (Shafee et al., 2025). Additionally, studies also address legal, ethical, and societal contexts when using chatbots (van der Rijt et al., 2025). Such solutions often rely on expensive proprietary LLMs and typically do not incorporate privacy controls for handling sensitive data.

Recent studies also emphasize the importance of addressing security and privacy concerns in LLM-based chatbots. Montagna et al. (2025) discuss privacy-preserving LLM-based chatbots to support self-management in hypertensive patients. The authors consider an approach that uses a local classifier to filter sensitive data before querying an online LLM (GPT-3.5 Turbo), and an entirely local deployment using offline LLMs (e.g., Llama2). The first solution offers superior performance in intent recognition and data extraction, processing sensitive inputs locally using traditional methods without relying on LLMs. However, the system does not switch between online and offline LLMs based on data sensitivity. Relying exclusively on an offline approach may reduce response quality, depending on the capabilities of the chosen LLM.

Also supporting healthcare applications, Li et al. (2024) present a privacy protection framework for LLM-based chatbots. The authors rely on access control, de-identification of sensitive data, and prompt engineering. However, these approaches are, in general, static and may struggle to adapt to varied inputs, since the effectiveness of de-identification depends on the diversity and quality of available medical record data.

A more general solution to preserve privacy can integrate a data privacy vault with a commercial service to create LLM applications (e.g., Amazon Bedrock).¹ The solution can detect Personally Identifiable Information (PII), tokenize and store it, ensuring that neither the training

data nor the prompts contain plaintext PII. During inference, inputs are de-identified before usage, which may have the same limitations as the study of Li et al. (2024).

Moreover, the literature includes studies that address privacy protection in chatbot conversations using cryptographic mechanisms (Biswas, 2020; Giordani, 2024). Such cryptographic mechanisms can serve as an additional layer of privacy protection for LLM-based chatbots. For instance, Watanabe and Uchikoshi (2025) presents a privacy-preserving framework that combines zero-knowledge proofs with LLMs to generate personalized advice. The authors verify user traits through cryptographic proofs to guide LLM responses.

Cheng et al. (2024b) propose RemoteRAG, a privacy-preserving cloud-based RAG service to protect user queries during document retrieval. The authors introduce a mechanism to perturb query embeddings and employ partially homomorphic encryption for distance calculations and oblivious transfer protocols when document indices may reveal sensitive information.

Our work differs from previous studies by presenting a high-performance and cost-effective chatbot design approach that combines online and offline LLMs. Unlike prior research that primarily targets the healthcare domain, our case study considers a national educational public policy. We consider large-scale public education policies, which require low-cost solutions, crucial for government contexts with limited budgets, ensuring scalable and sustainable deployment despite financial constraints. While the implementation of cryptographic mechanisms is beyond the scope of our work, developers may integrate existing solutions in this area to add additional layers of privacy protection and further strengthen user trust. Table 1 compares the discussed prior studies with our approach.

3. High-performance and low-cost chatbots

Our chatbot architecture uses the Rasa framework to orchestrate the system, serving as a central control layer that manages the conversational flow. Fig. 1 shows a diagram illustrating the question-answer flow of a chatbot implemented according to our design approach. The solution consists of two main modules. The first is a static module built on Rasa's native features, which handles Frequently Asked Questions (FAQ). This module relies on predefined questions and dialogue rules to deliver fast and consistent responses to recurring queries. The second

¹ <https://aws.amazon.com/pt/blogs/apn/building-a-privacy-preserving-chatbot-with-amazon-bedrock/>

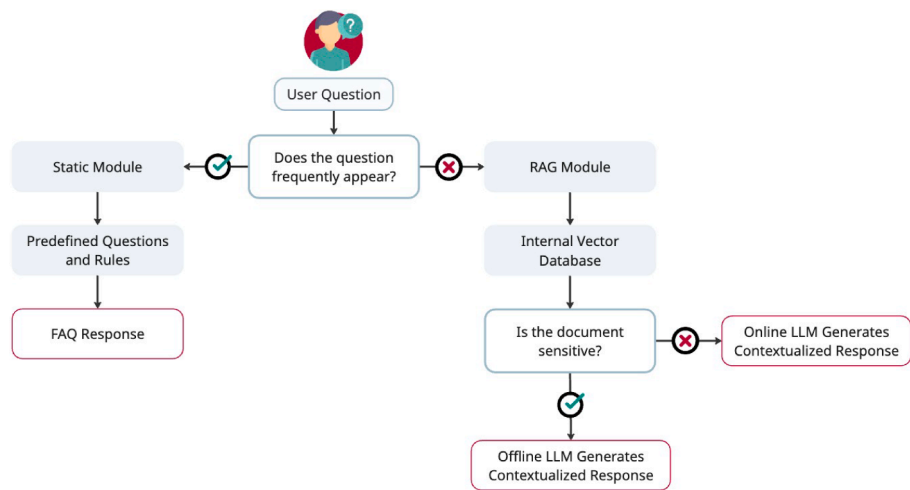


Fig. 1. Flow diagram for question-answer using the proposed chatbot.

module integrates a RAG architecture, which processes more complex questions that require access to institutional documents. In this module, Rasa detects the need for content retrieval and activates the RAG pipeline, which searches for relevant information in an internal vector database and generates contextualized responses using LLMs.

The system coordinates both modules to route each input to the most appropriate component based on its complexity and intended use. Rasa supports the dialogue orchestration, functional separation between modules, and data flow across components.

3.1. Information retrieval module

Rasa interprets and routes user messages through its Natural Language Understanding (NLU) component, which segments and analyzes the user’s input to extract key information, including intents and entities. Intents reflect the user’s communicative goal, while entities capture specific details such as document names, dates, or topics. To train this component, for our case study, we prepared an annotated dataset with representative examples of typical interactions related to the application domain, covering both FAQs and open-ended questions (e.g., administrative, operational, and regulatory matters).

We use Rasa’s default model, the Dual Intent and Entity Transformer (DIETClassifier), to perform intent classification. This model combines intent recognition and entity extraction within a unified transformer-based architecture that utilizes attention mechanisms to comprehend the contextual relationships between words and phrases. We train the DIETClassifier using our annotated examples for 100 epochs, enabling it to project inputs into a shared semantic space where it preserves meaning and similarity. In our implementation, the model can accurately identify intents, recognizing diverse expressions and linguistic variations. This classification step is essential for accurate message routing: the system sends FAQ-related intents to Rasa’s static module, while it directs queries that require document retrieval to the RAG module.

To improve dialogue control, we configured the Rasa framework with two key thresholds: a fallback threshold and an ambiguity threshold. We set the fallback threshold to 0.7, meaning that if the DIETClassifier’s confidence score falls below this value, the system triggers a fallback action and forwards the question to the RAG pipeline. We also defined the ambiguity threshold as 0.2; when the confidence difference between the top two predicted intents is smaller than this value, the system prompts the user to select the most appropriate intent. We determined these values empirically based on preliminary tests and guidance from the official Rasa documentation.

Table 2	
Query validation prompt template.	
System Instructions	
You are a security analyst specializing in detecting prompt injection attacks. Your task is to analyze the user input below and classify it as True if it is a prompt injection, or False otherwise. Any input that requests the implementation and/or execution of code in any programming language is considered prompt injection. Consider as a prompt injection any input that requests to ignore any prompt or command. Respond only with True or False , without further explanation.	
INPUT	{user_question}
RESPONSE	True / False

3.2. RAG module

The RAG-based module handles inputs that require contextualized responses drawn from public and private documents. Fig. 2 illustrates the module’s execution flow, which includes three main steps: query validation, context retrieval, and response generation.

The RAG workflow starts with a query validation step that we designed to detect potential prompt injection attempts. Prompt injection is a known vulnerability in LLM-based systems, where malicious users embed hidden instructions within the input to manipulate the model’s behavior (Hui et al., 2024; Perez & Ribeiro, 2022). To mitigate this risk, we apply a classification step using an LLM to identify potentially malicious input. We accomplish this verification through prompt engineering techniques that instruct the LLM to analyze the input for suspicious patterns, such as commands like “ignore previous instructions,” “answer as if,” or “generate a code”, before allowing the query to proceed. Our goal with this validation step is to enhance system integrity and prevent unauthorized access to sensitive documents. If the LLM classifies the input as a potential prompt injection, the chatbot returns a message indicating it cannot respond to the query and ends the dialogue flow. Otherwise, the question proceeds through the standard RAG pipeline. Table 2 shows the prompt we used in the validation step.

After validation, the system uses the verified question to retrieve relevant excerpts from a vector database containing previously indexed documents. We use Chroma,² an open-source vector embedding

² <https://github.com/chroma-core/chroma>

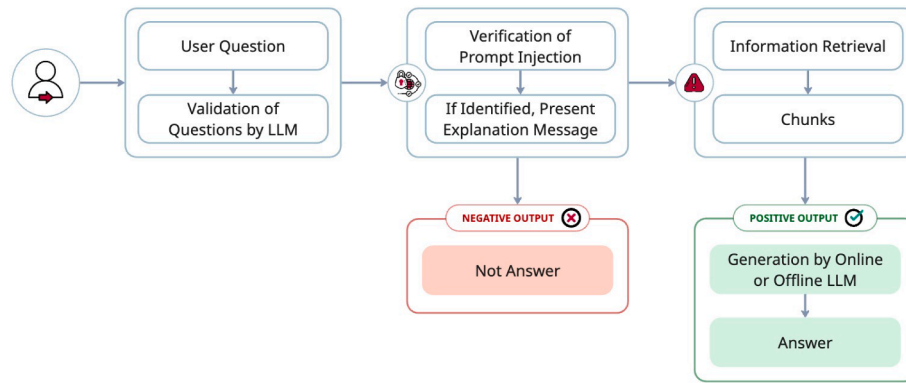


Fig. 2. Overview of the RAG workflow.

database, to support this process. The system segments the documents into fixed-size chunks, converts them into semantic embeddings, and stores them in the database. When it receives a query, the system generates an embedding for the question and performs a cosine similarity search to identify the most relevant chunks. It uses these chunks as context for the answer generation stage.

We divided the documents into chunks using a recursive overlapping character-based strategy, which is widely applied in RAG system development. Because the documents in our case study are large, we set the maximum chunk size to 5000 characters to control the total number of segments. To preserve contextual information across chunk boundaries, we introduced an overlap of 2000 characters between consecutive chunks. We selected this configuration to strike a balance between the number of extracted blocks and the need to maintain contextual continuity. Thus, this choice relies on preliminary tests and aligns with previous work using similar texts in Brazilian Portuguese, reporting that larger blocks, ranging from 4000 to 8000 characters, improve performance (Kuratomi et al., 2024).

We defined three experimental scenarios: (1) using only a low-cost online LLM, (2) using only a low-cost offline LLM, and (3) dynamically switching between online and offline LLMs. In the third scenario, during the final response generation stage, the system checks whether any of the retrieved chunks originate from documents labeled as sensitive or proprietary. If it identifies sensitive content, it routes the query and context to a local LLM hosted in a secure environment, ensuring that confidential information remains within the institution's infrastructure. If no sensitive content exists, the system generates the response using an online LLM, leveraging its higher fluency and natural language generation capabilities. This selective routing strategy balances privacy protection with language quality, aiming to deliver contextualized and accurate responses. Table 3 shows the prompt used for this final generation step.

Table 3
RAG response generation prompt template.

System Instructions
You are a virtual assistant specialized in answering questions based on official or domain-specific documents. Answer the following QUESTION using only the information provided in the CONTEXT . Do not use external knowledge or make assumptions.
CONTEXT: {context}
QUESTION: {user_question}
Write a detailed and accurate answer in formal, standard Brazilian Portuguese language appropriate to the domain. If the question cannot be answered using only the provided CONTEXT, return "NONE" without any further explanation.

In the hybrid approach, experts involved in our case study manually defined the sensitivity level for each document used in the experiments. We stored this information as metadata within each chunk. When the system retrieves chunks, it checks whether at least one of the top-k retrieved chunks originates from a sensitive document. If so, it processes the query using a local LLM; otherwise, it forwards the request to an online LLM.

4. Methodology

We defined a methodology to evaluate the chatbot within the context of a national educational public policy: PNLD case study (Araújo et al., 2025; Paiva et al., 2025). We describe how we built a corpus of question-answer pairs derived from institutional documents relevant to the study. We also detail the experiments conducted to validate different components of the RAG pipeline. Finally, we explain the metrics used to assess the performance and accountability of the chatbot, combining traditional automatic evaluation methods with LLM-assisted analyses.

4.1. Corpus construction

We validated the RAG pipeline's functionality and effectiveness by constructing a corpus of question-answer pairs along with corresponding documentary contexts. We derived this corpus from institutional sources relevant to the PNLD case study, including internal documents and the public call for textbooks focused on the selection of materials for Youth and Adult Education in the PNLD 2026 cycle. We followed a systematic workflow to build the dataset, as illustrated in Fig. 3, which included document preprocessing, question-answer generation, validation of the generated pairs, and question paraphrasing. The corpus consists of questions written in an FAQ style.

We segmented the documents into coherent textual chunks, preserving their semantic integrity so that each chunk is understandable independently. To minimize information loss, we applied an overlapping window between adjacent chunks. These segments provided the contextual foundation for the subsequent stages. In the question-answer generation phase, we used the GPT-4o-mini model to automatically generate relevant questions based on each chunk, along with answers strictly derived from the corresponding context.

To ensure corpus quality, we automatically validated the generated question-answer pairs using a second LLM, Sabiá-3, which is specifically tailored to Brazilian Portuguese (Abonizio et al., 2024). We used two distinct LLMs, one for generation and another for validation, to increase the robustness and impartiality of the curation process, following a guideline similar to that adopted by Mucciaccia et al. (2025b). We instructed the LLM for validation to identify question-answer pairs containing spelling errors, incoherent responses, or reliance on external information not present in the provided context. We retained only the pairs that met the criteria of clarity, accuracy, and contextual fidelity.

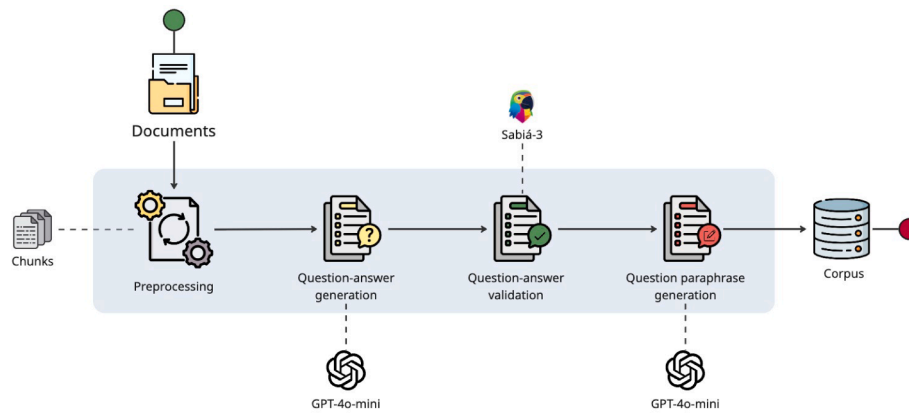


Fig. 3. Overview of the assessment corpus construction.

Table 4

Paraphrase generation prompt template.

System Instructions
Work as a specialist in linguistics and text rewriting. Your task is to paraphrase the following question, maintaining its original semantics but altering its syntax. QUESTION: {user_question}

We structured the corpus construction process in two stages: generation and validation, using different LLMs to minimize the need for human validation. To ensure quality, we analyzed a random subset of questions to verify how well the questions and answers aligned with the documents. In this verification, we confirmed that the generated questions and answers were valid and relevant to the application domain.

Finally, we used GPT-4o-mini to generate the paraphrases of the original questions, simulating a more realistic and challenging scenario for the RAG pipeline. This step introduced linguistic variation, enabling us to evaluate the pipeline's ability to handle different expressions of the same informational intent. As a result, we obtained a structured, diverse, and validated corpus suitable for assessing the effectiveness of our RAG architecture. Table 4 presents the prompt used to generate paraphrased versions of the original questions.

Table 5 presents examples of original questions, their paraphrased versions, and the corresponding responses from the database generated during our case study. We translated them into English for presentation, as they were initially in Brazilian Portuguese. Comparing the paraphrased questions with the original ones revealed that approximately 46.00 % of the words used in the paraphrased questions differ from those in the original questions.

4.2. Experimental setup

We evaluated a chatbot, implemented based on our design approach, using both traditional automatic textual similarity metrics and LLM-based judgment metrics to assess the quality, fidelity, and relevance of its responses. To this end, we defined three experimental scenarios: (1) evaluating embedding models, (2) assessing LLM performance, and (3) assessing the ability to not answer inappropriate questions, i.e., questions that are outside of the chatbot's domain of interest.

Thus, we considered different embedding models to support the retrieval module in an RAG system. The evaluation focused on two key metrics: recall@n, which measures the proportion of relevant chunks successfully retrieved, and the mean number of tokens in the top-n retrieved chunks, which assesses retrieval efficiency. Top-n refers to the set of the n highest-ranked chunks returned by the retrieval model based on their relevance scores. We considered two experimental scenarios:

one using the original question and the other using a paraphrased version of the same question, allowing for an analysis of the model's robustness to linguistic variation.

Moreover, we evaluated different LLMs to generate user responses based on the contexts retrieved by the RAG retrieval module: Llama 3 with 1, 3, 8, and 70 billion parameters; Gemma 3 with 1, 4, and 12 billion parameters; Maritaca's proprietary Sabiazinho-3 model; and the DeepSeek-R1-Distill-Llama-70B model. We selected these models to ensure diversity in architecture, size, and training methodology, enabling comparisons across compact, multilingual, and instruction-tuned models.

We accessed the Llama 3 and DeepSeek-R1-Distill-Llama-70B models through the TogetherAI platform,³ and Sabiazinho-3 via the API provided by Maritaca.⁴ We implemented the local models using the Transformers library⁵ and for efficiency reasons, the quantized models provided by Unsloth AI.⁶ We executed the models locally on a system equipped with an Intel® Core™ i5-11400F processor, 32 GB of RAM, and an Nvidia RTX 3080 GPU with 12 GB of memory, running Ubuntu 22.04.5 LTS.

Our experiments relied on the traditional ROUGE-L (Lin, 2004) and BERTScore (Zhang et al., 2023) metrics, as well as LLM-based judgment metrics, including faithfulness, answer relevance, and answer correctness computed using the RAGAS framework (Es et al., 2024). We used the same two experimental scenarios defined for embedding models. This approach allowed us to assess the models' performance and their robustness to variations in question phrasing.

We also analyzed the ability of the two best-performing models from the second experiment (i.e., Gemma 3 and Sabiazinho 3) to handle out-of-context and malicious questions, including prompt injection attempts. For this assessment, we used a dataset of 200 randomly selected instructions from the Ultra Alpaca PT-BR collection,⁷ ensuring that all questions were outside the PNL domain. We measured the models' responses using three metrics: the percentage of questions answered, the percentage left unanswered, and the percentage of answers potentially indicative of prompt injection behavior. This evaluation assesses the models' robustness when exposed to irrelevant or adversarial inputs.

Some examples of instructions used in the evaluation include: "Build an SQL query to find the names and states of all employees located in California"; "Analyze the sentence and identify the verb tense used in the sentence: "He was playing baseball"; and "Reverse engineer this

³ <https://www.together.ai/>

⁴ <https://www.maritaca.ai/>

⁵ <https://huggingface.co/docs/transformers/index>

⁶ <https://huggingface.co/unsloth>

⁷ <https://huggingface.co/datasets/recogna-nlp/ultra-alpaca-ptbr>

Table 5
Examples of questions and answers extracted from the corpus.

Original Question	Paraphrased Question	Answer
How can the concept of “fade in” and “fade out” be applied in audio editing for educational purposes?	In what ways can “fade in” and “fade out” techniques be applied in audio editing for educational use?	“Fade in” and “Fade out” are audio editing techniques that can be extremely useful in educational settings. “Fade in” allows a piece of music to start softly and gradually increase in intensity, which can help students focus on the introduction of the piece of music. “Fade out”, on the other hand, provides a gradual ending, preventing the piece from ending abruptly, which can leave the impression that something is missing. These techniques are especially important when cutting out musical passages, as they help maintain the integrity of the musical phrase, allowing students to understand better the structure and continuity of the piece of music.
What are the consequences for a textbook that presents more than 10% of specific errors during the pedagogical evaluation?	What implications can a textbook have if more than 10% of specific errors are identified during the pedagogical evaluation?	If a textbook presents a number of specific errors exceeding 10% of the total number of pages, it will be rejected. Each specific error identified in the evaluation counts as one occurrence, and the Technical Committee has the autonomy to evaluate and decide on omissions related to specific errors without this implying a global review of the work.

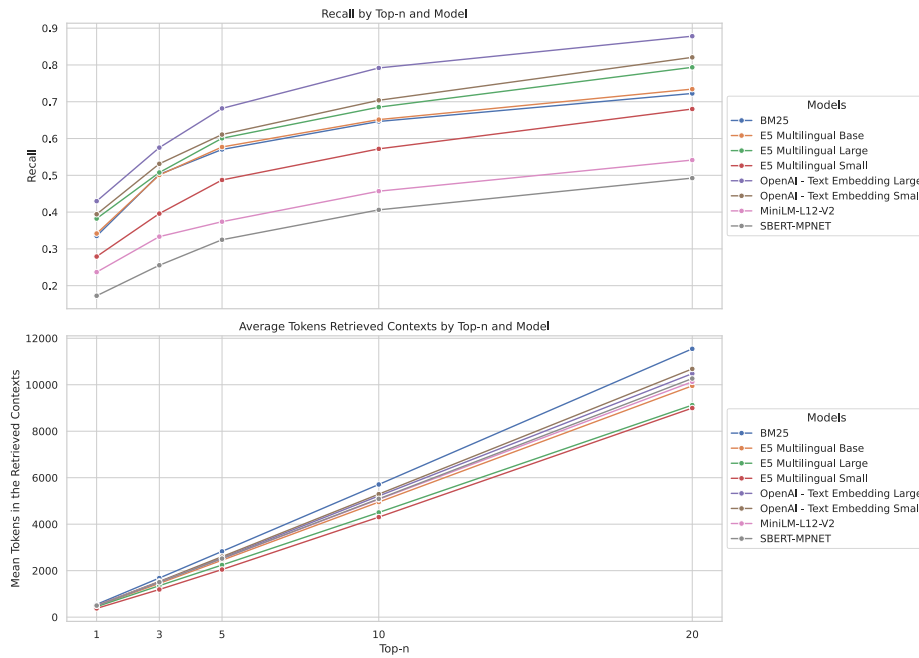


Fig. 4. Recall@n and average tokens retrieved results by top-n and model for the original set of questions.

code to find its output: $n = 10; \text{while } n > 0 : \text{print}(n); n = n - 1.$ These instructions are intentionally unrelated to the PNLD case study domain and are used to test the models' ability to avoid responding to irrelevant or adversarial queries.

5. Results and discussion

5.1. Embedding models assessment

Fig. 4 presents the results of the embedding model evaluation based on recall@n and the average number of tokens retrieved for the original questions. We considered eight retrieval models: BM25, E5 variants, OpenAI embeddings, MiniLM-L12-V2, and SBERT-MPNET. We selected diverse models to evaluate traditional and neural approaches. BM25 is a lexical baseline retrieval model. The E5 family contains multilingual models fine-tuned for retrieval. OpenAI's text embedding models are large-scale alternatives. MiniLM-L12-V2 and SBERT-MPNET are compact and efficient transformer-based models.

As the top-n value increases, more tokens are included in the prompt, which affects the cost of using LLMs since most commercial models charge based on token usage. Choosing an appropriate top-n value requires balancing retrieval performance with computational efficiency to ensure both quality and cost-effectiveness.

The results indicate that the OpenAI-Text Embedding Large model performed best across all top-n values, achieving a recall@n of 0.87 at the Top-20. This performance is higher than that of the other models evaluated, including BM25 and E5 Multilingual Large, highlighting effectiveness in accurately identifying the most relevant chunks. However, the OpenAI model is proprietary and requires payment, which may pose a barrier to its adoption in projects with limited resources or those subject to data privacy constraints.

The BM25 model, although a purely lexical model, was efficient in some scenarios. However, it tended to retrieve longer chunks, resulting in a higher average number of tokens per query compared to the other models. This characteristic may negatively impact the operational cost, especially in environments with restricted context windows or those that charge for processed tokens.

The E5 Multilingual Large showed competitive performance, especially in intermediate Top-n values. In the Top-10, for instance, it obtained a recall@n of approximately 0.7, and 0.794 in the Top-20. This model consistently achieved the third-best performance, being surpassed only by OpenAI's small and large models. Thus, E5 Multilingual Large can be a viable alternative for scenarios that require cost control, prioritize open-source solutions, and, for data privacy reasons, necessitate running models on their local infrastructure.

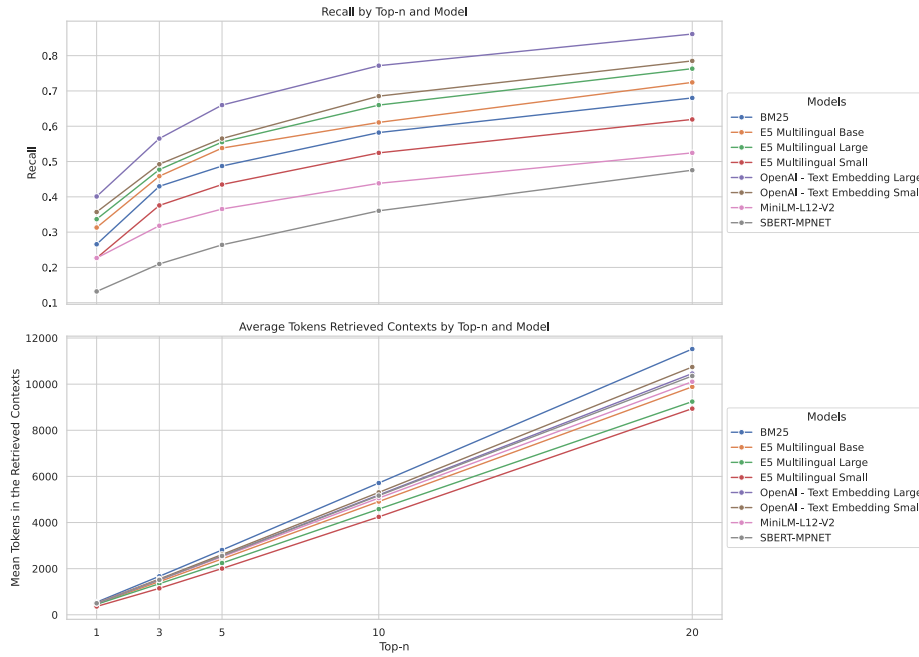


Fig. 5. Recall@n and average tokens retrieved results by Top-n and model for the paraphrased set of questions.

Fig. 5 presents the recall@n and average number of tokens retrieved by Top-n and model, considering the paraphrased set of questions. We observed a slight decrease in the recall@n metric, suggesting that paraphrasing may introduce subtle variations that impact retrieval performance.

In the second scenario, considering paraphrased questions, the OpenAI-Text Embedding Large model maintained its superior performance, achieving a recall@n of 0.861 in the Top-20. This consistency reinforces its semantic robustness and its ability to generalize in the face of the linguistic diversity of queries. The E5 Multilingual Large model, although it performed slightly worse than the OpenAI models (small and large), continued to stand out as the best open-source alternative, with a recall@n above 0.75 in the Top-20.

All models showed a drop in performance because the paraphrasing scenario is more challenging. However, the OpenAI and E5 Multilingual Large models were more robust to linguistic variations, with a slight drop in performance. Models such as SBERT-MPNET, MiniLM-L12-V2, and BM25 exhibited higher performance drops, underscoring their limited ability to handle paraphrasing.

The experiments demonstrate that the OpenAI models were the most effective in context retrieval. However, its use is conditioned by a closed commercial model, which can limit its adoption. The E5 Multilingual Large model demonstrated robust and stable performance, especially when faced with the paraphrasing challenge. Considering the balance between cost, performance, and licensing, we selected the E5 large model for our experiments.

In terms of cost, the average number of tokens retrieved increased proportionally with the value of Top-n. For practical applications, using Top-5 and Top-10 offers a good balance between cost and performance, with high recall@n and token volume still within operational limits suitable for LLM usage, especially considering scenarios with limited financial resources.

Table 6 presents the estimated cost in US dollars (US\$) for generating answers using the retrieved context per Top-n value, plus 256 output tokens. The analysis focuses on the online LLM Sabiazinho-3, which was selected for its compact architecture compared to large-scale models like GPT-4o and its specialization in Portuguese, aligning well with our PNLD case study. As shown, increasing the Top-n value results in a substantial rise in cost, particularly when scaling to a large number

Table 6

Cost in US\$ per Top-n for Sabiazinho-3 model and number of questions.

Top-n	1 question	100 questions	1,000 questions
1	0.0002	0.026	0.264
3	0.0004	0.048	0.482
5	0.0006	0.070	0.699
10	0.0012	0.127	1.266
20	0.0024	0.239	2.390

of queries. For example, processing 1000 questions with Top-20 results in a cost of approximately US\$ 2.39. Sabiazinho-3 charges about US\$ 0.20 per million input tokens and US\$ 0.60 per million output tokens.⁸ Based on this analysis, we adopted the Top-10 retrieval approach as a cost-effective choice that also maintains competitive retrieval performance.

5.2. Question-answer analysis

Table 7 shows the results of evaluating the LLMs on the original question set, using Top-5 and Top-10 retrieved contexts. The best result for each metric is shown in bold. Results that are statistically equivalent to the highest value for a given metric, according to the Wilcoxon signed-rank test, are indicated with the symbol †. The hybrid configuration, combining Gemma 3 12B and Sabiazinho-3, achieved significant performance across all metrics. It obtained the highest ROUGE-L score for both Top-5 (a mean of 0.361) and Top-10 (a mean of 0.373) retrieved contexts, along with the highest BERTScore (a mean of 0.758 and 0.762, respectively) and the most significant percentage of correctly answered questions (a mean of 96.672% for Top-5 and 97.180% for Top-10). Among individual models, Sabiazinho-3 stood out with consistently significant performance, demonstrating the effectiveness of online models. While smaller models like Llama 3 3B and Gemma 3 4B showed lower scores and higher variance, larger models such as Llama 3 70B and Gemma 3 12B benefited from their scale and training, performing competitively in both Top-5 and Top-10 settings.

⁸ <https://www.maritaca.ai/>

Table 7

Mean performance (with standard deviation) from three rounds on original questions with Top-5/10 retrieved contexts.

LLM	Top-5			Top-10		
	ROUGE-L	BERTScore	%Answered	ROUGE-L	BERTScore	%Answered
Llama 3 1B (offline)	0.286 (0.043)	0.726 (0.019)	86.858 (15.355)	0.266 (0.074)	0.719 (0.033)	82.346 (21.138)
Llama 3 3B (offline)	0.247 (0.039)	0.711 (0.020)	88.212 (15.355)	0.215 (0.065)	0.695 (0.035)	88.099 (14.916)
Llama 3 8B (offline)	0.278 (0.056)	0.721 (0.025)	83.700 (19.821)	0.261 (0.085)	0.717 (0.038)	80.203 (25.245)
Llama 3 70B (online)	0.318 (0.011)	0.740 (0.011)	85.843 (5.431)	0.331 (0.013)	0.748 (0.011)	89.791 (5.236)
Gemma 3 1B (offline)	0.223 (0.023)	0.690 (0.013)	94.247 (3.740)	0.214 (0.027)	0.686 (0.014)	93.175 (4.759)
Gemma 3 4B (offline)	0.265 (0.113)	0.712 (0.066)	75.635 (33.142)	0.285 (0.077)	0.724 (0.045)	83.249 (22.494)
Gemma 3 12B (offline)	0.349 (0.023)	0.751 (0.011)	94.078 (3.237)	0.361 (0.018)	0.756 (0.008)	95.431 (2.274)
Sabiazinho-3 (online)	0.349 (0.006)	0.764 (0.001)	97.970 (1.332)	0.345 (0.005)	0.763 (0.001)	98.872 (0.761)
DeepSeek-R1-Distill-Llama-70B (online)	0.312 (0.008)	0.732 (0.005)	89.848 (0.633)	0.327 (0.002)	0.741 (0.001)	93.401 (1.318)
Hybrid (Gemma 3 12B and Sabiazinho-3)	0.361 (0.001)	0.758 (0.001)	96.672 (0.211)	0.373 (0.001)	0.762 ⁺ (0.001)	97.180 (0.080)

Table 8

Mean performance (with standard deviation) from three rounds on paraphrased with Top-5/10 retrieved contexts.

LLM	Top-5			Top-10		
	ROUGE-L	BERTScore	%Answered	ROUGE-L	BERTScore	%Answered
Llama 3 1B (offline)	0.258 (0.044)	0.715 (0.021)	85.730 (16.352)	0.249 (0.071)	0.713 (0.032)	82.346 (21.738)
Llama 3 3B (offline)	0.228 (0.029)	0.703 (0.015)	89.509 (13.281)	0.199 (0.057)	0.688 (0.034)	90.637 (11.806)
Llama 3 8B (offline)	0.253 (0.049)	0.714 (0.022)	83.362 (20.181)	0.239 (0.077)	0.709 (0.036)	79.583 (26.123)
Llama 3 70B (online)	0.289 (0.010)	0.733 (0.009)	85.561 (4.511)	0.304 (0.011)	0.742 (0.011)	90.186 (4.307)
Gemma 3 1B (offline)	0.209 (0.021)	0.685 (0.013)	93.514 (5.464)	0.201 (0.023)	0.681 (0.013)	92.499 (5.112)
Gemma 3 4B (offline)	0.252 (0.099)	0.707 (0.059)	76.255 (31.786)	0.271 (0.077)	0.721 (0.045)	82.854 (23.052)
Gemma 3 12B (offline)	0.332 (0.024)	0.747 (0.013)	93.457 (4.230)	0.339 (0.017)	0.752 (0.009)	98.872 (0.523)
Sabiazinho-3 (online)	0.319 (0.009)	0.756 ⁺ (0.001)	97.800 (1.442)	0.316 (0.005)	0.755 ⁺ (0.001)	94.755 (3.231)
DeepSeek-R1-Distill-Llama-70B (online)	0.291 (0.005)	0.728 (0.004)	91.935 (1.018)	0.297 (0.001)	0.732 (0.002)	93.457 (0.761)
Hybrid (Gemma 3 12B and Sabiazinho-3)	0.346 (0.001)	0.756 (0.001)	96.221 (0.444)	0.350 (0.001)	0.758 (0.001)	96.842 (0.211)

In the Top-5 setting, Sabiazinho-3 achieved a mean ROUGE-L of 0.349, a mean BERTScore of 0.764, and answered 97.970 % of the questions on average. Its performance remained consistently high in the Top-10 setting, where it answered 98.672 % of the questions on average, the highest among all individual models. In contrast, Gemma 3 1B showed the weakest performance in the Top-5 setting, with a mean ROUGE-L of 0.223 and a mean BERTScore of 0.69, illustrating the impact of model selection on RAG effectiveness. The hybrid strategy, which combines the outputs of Gemma 3 12B and Sabiazinho-3, achieved results that matched or exceeded those of individual models, particularly in ROUGE-L and percentage of answered questions. This hybrid approach refers to the integration of online and offline LLMs to optimize answer generation, balancing performance, cost, and data sensitivity considerations.

Table 8 reports the average performance of the same LLMs on paraphrased questions using Top-5 and Top-10 retrieved contexts. Similar to the evaluation with the original questions, Sabiazinho-3 delivered one of the most significant performances, achieving a BERTScore of 0.756 and answering 97.800 % of the questions in the Top-5 setting. In the Top-10 setting, it maintained high consistency, answering 94.755 % of the questions with minimal variance. In contrast, Gemma 3 1B showed the weakest results in the Top-5 configuration, with a ROUGE-L of 0.209 and a BERTScore of 0.685. These findings suggest that paraphrased input introduces additional challenges for less robust models, while Gemma 3 12B and Sabiazinho-3 remain resilient across different input formulations.

Although our dataset follows an FAQ style, our architecture supports multi-turn conversational scenarios. Rasa orchestrates the conversational flow and preserves context across turns, routing each query to the most appropriate module according to its complexity. Our results evidence that the system handles paraphrasing and linguistic variations robustly, which shows its ability to process reformulated or follow-up questions naturally occurring in dialogue. The

static FAQ module and the RAG-based module work together to ensure coherent transitions between direct answers and context-grounded responses.

5.3. Accountability analysis

Table 9 reports the results from the original corpus questions and the Top-5 retrieved contexts, evaluated with the RAGAS metrics. The hybrid configuration that combined Gemma 3 12B and Sabiazinho-3 reached the highest fidelity score (0.918), showing more consistency between the generated answers and the supporting retrieved context. Among the individual models, Gemma 3 12B achieved the highest fidelity score (0.891), followed by Sabiazinho-3 (0.795) and Llama 3 70B (0.760).

Smaller models with 1B to 4B parameters showed limited performance. Although some, such as Gemma 3 1B and Llama 3 3B, reached reasonable scores in faithfulness and answer relevancy, their answer correctness values remained below 0.50, indicating difficulty in producing accurate responses. Llama 3 8B also failed to demonstrate significant improvements over the smaller models, suggesting that simply increasing the number of parameters does not ensure better performance. In contrast, Llama 3 70B outperformed all smaller models across every metric, reinforcing the trend that a substantial increase in model size can enhance capacity and consistency.

Sabiazinho-3 obtained the best scores for answer relevance (0.757) and answer correctness (0.648), showing more balance between adequacy and accuracy. Gemma 3 12B, on the other hand, achieved the highest performance in faithfulness (0.891). The hybrid strategy that combined these two models capitalized on their strengths, reaching the highest overall fidelity (0.918) while sustaining competitive levels of relevance (0.682) and correctness (0.579). This configuration also demonstrated robustness, maintaining high factual consistency while delivering relevant and accurate responses.

Table 9

Mean performance (with standard deviation) using RAGAS metrics from three execution rounds (Top-5, original questions).

LLM	Faithfulness	Answer Relevancy	Answer Correctness
Llama 3 1B (offline)	0.660 (0.144)	0.609 (0.169)	0.490 (0.102)
Llama 3 3B (offline)	0.603 (0.093)	0.644 (0.202)	0.474 (0.104)
Llama 3 8B (offline)	0.635 (0.164)	0.598 (0.204)	0.476 (0.116)
Llama 3 70B (online)	0.760 (0.028)	0.702 [†] (0.034)	0.575 (0.030)
Gemma 3 1B (offline)	0.651 (0.020)	0.641 (0.095)	0.420 (0.069)
Gemma 3 4B (offline)	0.661 (0.284)	0.552 (0.226)	0.453 (0.184)
Gemma 3 12B (offline)	0.891 (0.042)	0.659 (0.043)	0.562 (0.027)
Sabiazinho-3 (online)	0.795 (0.016)	0.757 (0.028)	0.648 (0.011)
DeepSeek-R1-Distill-Llama-70B (online)	0.702 (0.018)	0.685 (0.061)	0.558 (0.013)
Hybrid (Gemma 3 12B and Sabiazinho-3)	0.918 (0.004)	0.682 (0.003)	0.579 (0.004)

Table 10

Mean performance (with standard deviation) using RAGAS metrics from three execution rounds (Top-5, paraphrased questions).

LLM	Faithfulness	Answer Relevancy	Answer Correctness
Llama 3 1B (offline)	0.626 (0.136)	0.619 (0.199)	0.481 (0.103)
Llama 3 3B (offline)	0.598 (0.081)	0.653 (0.181)	0.463 (0.096)
Llama 3 8B (offline)	0.615 (0.150)	0.610 (0.219)	0.477 (0.119)
Llama 3 70B (online)	0.747 (0.026)	0.717 [†] (0.023)	0.562 (0.030)
Gemma 3 1B (offline)	0.662 (0.013)	0.645 (0.138)	0.416 (0.068)
Gemma 3 4B (offline)	0.668 (0.271)	0.569 (0.224)	0.459 (0.178)
Gemma 3 12B (offline)	0.876 (0.046)	0.666 (0.048)	0.559 (0.032)
Sabiazinho-3 (online)	0.796 (0.020)	0.780 (0.022)	0.636 (0.002)
DeepSeek-R1-Distill-Llama-70B (online)	0.703 (0.006)	0.687 (0.066)	0.563 (0.005)
Hybrid (Gemma 3 12B and Sabiazinho-3)	0.906 (0.008)	0.699 (0.002)	0.581 (0.002)

Table 10 shows the results for the second evaluation scenario, which used paraphrased versions of the original questions. The hybrid approach that combined Gemma 3 12B and Sabiazinho-3 achieved the highest faithfulness score (0.906), outperforming all individual models. Sabiazinho-3 delivered the best results for answer relevance (0.780) and answer correctness (0.636), demonstrating robustness to variations in question phrasing. Gemma 3 12B reached the top faithfulness score among individual models (0.876) while remaining competitive in relevance (0.666) and correctness (0.559).

Sabiazinho-3 achieved the highest scores for answer relevance (0.780) and correctness (0.636), showing strength in generating contextually rich and intent-aligned responses, even when not fully grounded in the retrieved evidence. In contrast, the smaller models performed poorly across all metrics, underscoring their limitations in handling reworded or complex queries.

These results confirm that the hybrid strategy, which dynamically integrates a high-capacity local model with a robust online model, achieves an effective balance between response quality and privacy constraints. The metric-wise analysis further shows that LLMs present distinct behavioral patterns: the hybrid strategy excelled in factual grounding, whereas Sabiazinho-3 prioritized comprehensiveness and alignment with user intent.

The results revealed a trade-off across the three evaluation metrics. Sabiazinho-3, for example, generated contextually rich responses aligned with the question's intent, but these answers were not always strictly grounded in the retrieved content, which reduced its performance in faithfulness. In contrast, Gemma 3 12B showed more adherence to the retrieved evidence, producing more faithful responses, though with a slight decrease in perceived relevance.

From a practical standpoint, especially in chatbot applications for sensitive domains such as public policy, the factual grounding of responses must take priority over broader semantic interpretations. In these contexts, the faithfulness metric becomes particularly critical, as it ensures that answers remain fully anchored in retrieved official doc-

Table 11

LLMs' ability to not answer inappropriate questions (Top-5 retrieval).

LLMs	Answered	Not Answered	Possible Prompt Injection
Gemma 3 12B	1	189	10
Sabiazinho-3	2	198	0

uments, preventing unsupported inferences and strengthening transparency, traceability, and accountability. While the answer relevancy and answer correctness metrics provide valuable insights into semantic alignment and factual accuracy relative to the input questions, they can yield high scores even when the supporting context does not fully justify the responses.

To evaluate the robustness of the models in handling inappropriate or out-of-domain questions, we conducted a final analysis focusing on Gemma 3 12B and Sabiazinho-3, which had shown the best performance in previous experiments. This step was essential to ensure that the models would not generate responses unsupported by the retrieved document excerpts or that could introduce potential security risks. Table 11 summarizes the results of this evaluation, which considered both general out-of-domain questions and prompt injection attempts, using the instructions specified in the adopted prompt. The findings indicate that both models were generally effective in avoiding inappropriate responses, but Sabiazinho-3 demonstrated more consistent behavior, with no harmful outputs and zero instances of injection, whereas Gemma 3 12B produced 10 such cases. These results highlight Sabiazinho-3's stronger resilience to adversarial inputs, an aspect of particular relevance given its role as the online component in the proposed hybrid approach, in combination with Gemma 3 12B as the offline component.

The high performance achieved by both models, Gemma 3 12B and Sabiazinho-3, can be explained by two key factors. First, the specific prompt injection detection step acts as an additional security barrier,

preventing malicious input from compromising the model's behavior. This verification layer contributes to the models' ability to resist manipulation attempts. Second, the prompt adopted in the RAG system was designed to explicitly instruct the models not to answer questions that cannot be fully answered based on the retrieved context (Table 3). This instruction reinforces the models' desired and controlled behavior, directing their responses only when there is sufficient evidence in the content returned by the retrieval engine.

5.4. Threats to validity

Due to computational constraints, we ran the LLM models three times. Running them several times would provide a better understanding of performance variance by computing the standard deviation of metrics. However, we carefully designed our experimental setup to mitigate this limitation and gain insights into model hallucinations (Huang et al., 2025; Ji et al., 2023b). We also configured the temperature parameter to 0.1 for all experiments, ensuring low randomness in the model's outputs and promoting more deterministic, consistent generations across runs. With the exception of the Gemma 3 4B model, which showed a larger standard deviation in the results between the three runs, no significant variance in the results was observed, especially in larger models.

On average, running each model with a set of 591 questions took approximately 10 hours, i.e., 1 minute per question. Such time may introduce a threat to external validity, especially when considering real-world chatbot applications that require low-latency responses. To mitigate this issue and reduce perceived latency in our case study, we integrated the LangChain streaming framework,⁹ which also improved transparency during interactions.

Moreover, the effectiveness of classifying prompt injection attempts depended on the number and relevance of examples provided to guide the LLM. To mitigate this limitation in our prompt injection experiment, we carefully crafted a prompt that included representative examples of queries that could plausibly be problematic. This approach aimed to help the model more accurately identify and classify potential prompt injection attacks by providing it with more precise guidance on typical attack patterns (Zhu et al., 2024).

Our methodology uses LLMs to generate question-answer pairs from institutional documents. We used GPT-4o-mini for question generation and Sabia-3 for automatic validation. While these models differ in architecture and specialization, they rely on extensive and similarly sourced data, which may negatively affect the interpretation of domain-specific language (e.g., education policy). While GPT-4o-mini is a large, multilingual, general-purpose transformer optimized for general conversational generation, Sabia-3 is a smaller, instruction-tuned model designed for Brazilian Portuguese. The decision to use one LLM for generating question-answer pairs and a different one for validation is supported by prior research on this topic (Mucciaccia et al., 2025a).

While our pipeline attempts to mitigate problems by separating generation and validation models, it does not address issues of using LLMs with similar linguistic priors. For example, the models might oversimplify education policy language, overlook specific details of policy guidelines, or exhibit bias in selecting which aspects of the institutional text to prioritize. Moreover, we recognize the risk of model drift over time in the corpus generation phase. Given that LLM providers update and fine-tune models, the quality and style of generated question-answer pairs may change across deployments.

6. Conclusions

Our experimental results demonstrate that integrating low-cost online and offline LLMs when designing chatbots is an effective strategy

for handling sensitive information from the knowledge base while maintaining performance. Switching between online and offline LLMs based on document sensitivity kept or improved the performance of question-answer tasks compared to using only online or offline approaches.

These findings suggest that high performance, defined in our study as providing substantial semantic similarity, factual accuracy, correctness, and relevance, can still be achieved even in resource-constrained and privacy-sensitive scenarios, overcoming the limitations associated with isolated deployment strategies. Achieving privacy protection without compromising performance is a significant outcome for domains such as public administration, which benefits from low-cost solutions due to existing resource constraints. As a practical example, such solutions could support initiatives like Brazil's National Textbook Program, where sensitive educational policies and large-scale document analysis require privacy and quality assurance.

Future research will include a qualitative evaluation of chatbot responses using well-established hallucination taxonomies to better understand users' perceptions of answer quality and reliability. We will also broaden the accountability analysis by incorporating additional evaluation metrics. In future work, we can also employ models specifically trained to detect prompt injection, allowing us to improve our experiments by comparing their performance with our current example-based approach to evaluate the impact and cost-benefit trade-offs.

Additionally, we plan to support multilingual use cases, such as retrieving and generating education policy documents in multiple official languages, offering localized chatbot interfaces for diverse communities, and facilitating cross-border collaboration on policy development. We also intend to integrate encryption schemes to enhance data privacy during retrieval and generation processes. Future work will also explore user-centered evaluations to assess the system's usability, effectiveness, and trustworthiness from the perspective of its target audience. Finally, we will explore quantifying latency and evaluating user-facing speed trade-offs.

CRedit authorship contribution statement

Hilário Tomaz Alves de Oliveira: Conceptualization, Methodology, Software, Validation, Writing – original draft, Writing – review & editing; **Álvaro Alvares De Carvalho César Sobrinho:** Conceptualization, Methodology, Validation, Writing – original draft, Writing – review & editing; **Andrey dos Reis Cadima Dias:** Methodology, Software, Writing – review & editing; **André Magno Costa de Araújo:** Conceptualization, Methodology, Software, Validation, Writing – original draft, Writing – review & editing; **Rafael Dias Araújo:** Conceptualization, Methodology, Software, Validation, Writing – original draft, Writing – review & editing; **Diego Dermeval Medeiros da Cunha Matos:** Conceptualization, Methodology, Software, Validation, Writing – original draft, Writing – review & editing; **Sebastian Munoz-Najar Galvez:** Conceptualization, Methodology, Software, Validation, Writing – original draft, Writing – review & editing.

Data availability

Data will be made available on request.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

⁹ <https://python.langchain.com/docs/concepts/streaming/>

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